



COMSATS University Islamabad, Attock Campus

Department of Computer Science

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Assignment 4

TASK 1 SS

```
1 # Small primes
2 p, q = 11, 13
3 n = p * q
4 phi = (p-1)*(q-1)
5
6 e = 7 # public key
7
8 # Find d (private key)
9 def inv(e, phi):
10     for i in range(1, phi):
11         if (e*i) % phi == 1:
12             return i
13
14 d = inv(e, phi)
15
16 # Encrypt / Decrypt
17 def enc(m): return pow(m, e, n)
18 def dec(c): return pow(c, d, n)
19
20 # Test
21 msg = [ord(c) for c in "Ali"]
22 cipher = [enc(x) for x in msg]
23 plain = ''.join(chr(dec(x)) for x in cipher)
24
25 print("Encrypted:", cipher)
26 print("Decrypted:", plain)
```

OUTPUT:

```
PS C:\Users\zakar\OneDrive\Desktop\TEAM ASSIGNMENT4> & C:\Users\zakar\AppData\Local\Programs\Python\Python314\python.exe "C:\Users\zakar\OneDrive\Desktop\TEAM ASSIGNMENT4\TEAM_1.py"
Encrypted: [85, 4, 118]
Decrypted: All
PS C:\Users\zakar\OneDrive\Desktop\TEAM ASSIGNMENT4>
```

TASK 2 SS

```
1 import hashlib
2
3 # Hash
4 def hash_msg(m):
5     return int(hashlib.sha256(m.encode()).hexdigest(), 16)
6
7 # Sign
8 def sign(m):
9     h = hash_msg(m)
10    return pow(h, g, n)
11
12 # Verify
13 def verify(m, sig):
14     h = hash_msg(m)
15     return pow(sig, e, n) == h & n
16
17 # Test
18 message = "Hello"
19 signature = sign(message)
20
21 print("Signature:", signature)
22 print("Valid:", verify(message, signature))
23
24 # Modify message
25 fake = "Hella"
26 print("After change:", verify(fake, signature))
```